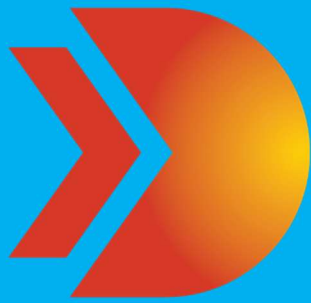




**Dhee
Coding
Lab**

JDBC [JAVA DATABASE CONNECTIVITY]



PUNITH B



API [Application Programming Interface]

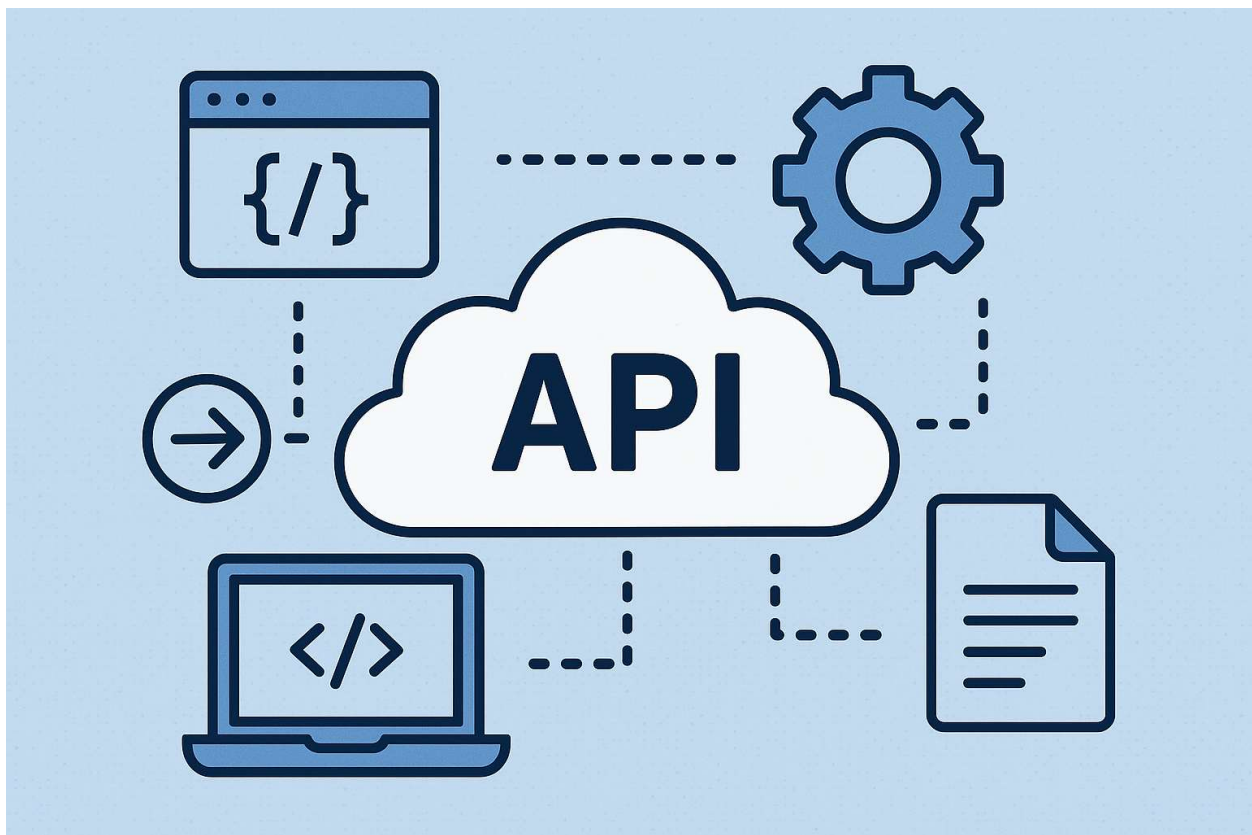
- API is used to establish connection between one application and another application.
- API is a set of rules that allows one piece of software or system to communicate with another.
- The result of the API is loose coupling.

Note: Loose coupling means reducing the dependencies between two components i.e. changing in the implementation will not affect the user.

The back bone of API is Abstraction

The process of displaying only the necessary or essential data by hiding the non-essential data to the end user is known as Abstraction.

An API (Application Programming Interface) is basically a contract between the provider and the user. The user doesn't need to know how something is implemented, only how to use it.

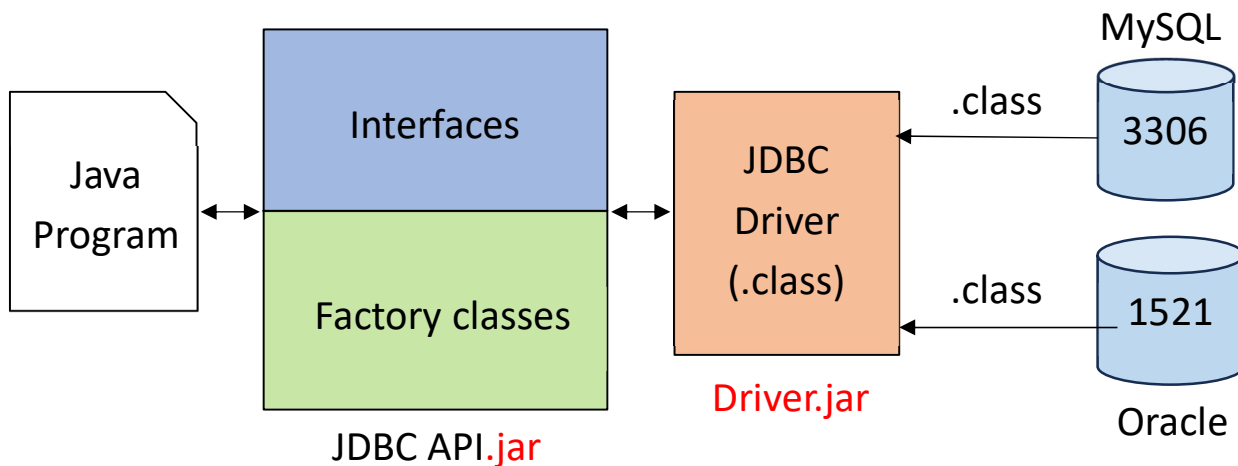




JDBC API

JDBC API is used to establish connection between Java and Database.

Architecture of JDBC API



- JDBC API is given by Sun Microsystems and is available in JRE library in the form of jar file.
- JDBC API consists of two packages. They are:
 1. java.sql
 2. javax.sql
- JDBC API consists of Interfaces and Factory classes.
- Implementation classes will be given by respective database servers or vendors.
- These implementation classes given by database servers are known as JDBC Drivers.

Interfaces available in JDBC API

- **Connection:** This interface is used to establish connection between Java and database.
- **Statement:** This interface is used to create a platform for executing the SQL Queries.
- **PreparedStatement:** This interface extends Statement Interface and also used to create a platform for executing SQL queries.
- **ResultSet:** This interface is used to retrieve the data from the resultant table that will be stored in cursor or buffer memory.



Factory class present in JDBC API

The only factory class present in JDBC API is known as **DriverManager**. Basically, it is a utility class which is responsible for managing the driver classes(JDBC Driver) given by the database servers or vendors.

Note: All these Interfaces and factory classes are available in java.sql package.

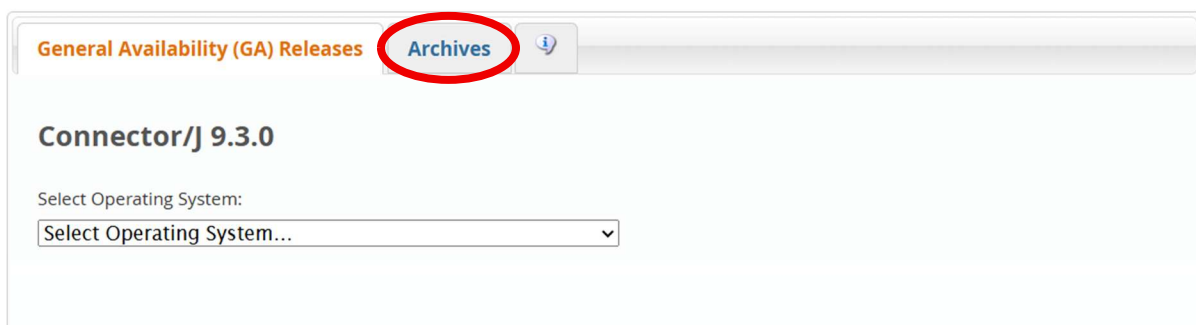
JDBC Driver / implementation class given by the Database servers or vendors

Oracle : OracleDriver Fully Qualified Class name: “oracle.jdbc.driver.OracleDriver”
MySQL : Driver Fully Qualified Class name: “com.mysql.cj.jdbc.Driver”

Note: As our DB vendor is MySQL, we need to download the mysql-connector.jar file from MySQL website.

Procedure to download mysql-connector.jar file

- Open web browser and search for mysql connector-j download. or Visit <https://dev.mysql.com/downloads/connector/j/>
- Choose Archives in the below tab.



- Choose Product version as 8.0.33 and Operating System as Platform Independent.
- Two download options will be available. Choose Platform Independent (Architecture Independent) ZIP archive [4.8MB]



MySQL Product Archives

MySQL Connector/J (Archived Versions)

Please note that these are old versions. New releases will have recent bug fixes and features!
To download the latest release of MySQL Connector/J, please visit MySQL Downloads.

Product Version: 8.0.33
Operating System: Platform Independent

Platform Independent (Architecture Independent), Compressed TAR Archive	Mar 7, 2023	4.0M	Download
(mysql-connector-j-8.0.33.tar.gz)		MD5: 7ada9973f9e636669b59790c76b52371	
Platform Independent (Architecture Independent), ZIP Archive	Mar 7, 2023	4.8M	Download
(mysql-connector-j-8.0.33.zip)		MD5: b637bc88f3b01d6f7bb322f01c083000 Signature	

We suggest that you use the MD5 checksums and GnuPG signatures to verify the integrity of the packages you download.

MySQL open source software is provided under the GPL License.

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- After downloading the ZIP file, extract it. A folder will be extracted with a name **mysql-connector-j-8.0.33**.
- Open the file, a **mysql-connector-j-8.0.33.jar** file is appeared. That is what the implementations given by MySQL.
- Further steps is provided in steps of JDBC concept.

Name	Date modified	Type	Size
Earlier this week			
build.xml	6/30/2025 9:19 AM	Microsoft Edge HT...	95 KB
CHANGES	6/30/2025 9:19 AM	File	274 KB
INFO_BIN	6/30/2025 9:19 AM	File	1 KB
INFO_SRC	6/30/2025 9:19 AM	File	1 KB
LICENSE	6/30/2025 9:19 AM	File	70 KB
mysql-connector-j-8.0.33.jar	6/30/2025 9:19 AM	Executable Jar File	2,424 KB
README	6/30/2025 9:19 AM	File	2 KB
src	6/30/2025 9:19 AM	File folder	